
Image Plagiarism Detection with Equivariant Encoders

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Abstract

A pirated image is rarely a pixel-perfect copy: rotations, reflections, crops, and re-compression are routine, and a detector should map all of them to the same point in feature space. We ask whether building these symmetries into the encoder of a siamese plagiarism detector is preferable to approximating them with data augmentation, and we resolve the question separately for the dihedral group of the square and for the special Euclidean group of the plane. Three encoders are compared inside one pipeline: a non-equivariant ViT-Tiny baseline trained with group-specific geometric augmentation, an Octic ViT with exact dihedral invariance, and a Harmformer with continuous roto-translational invariance. Training pairs are formed on COCO with a cross-batch protocol; evaluation uses cross-domain pairs from DomainNet, in an overall regime under the training-time augmentation distribution and a per-stratum regime that isolates one geometric transformation at a time, both at a fixed threshold. For the dihedral group, the architecturally invariant Octic ViT trained without geometric augmentation matches the augmented baseline on every dihedral transformation and roughly triples its precision on negatives, which supports the equivariance hypothesis. For the special Euclidean group, the verdict is mixed: the Harmformer improves overall metrics over the augmented baseline and is exact on right-angle rotations and reflections, but collapses on off-axis rotations and on translation, which we attribute to interpolation noise at low input resolution rather than to the equivariance principle.

Keywords: image plagiarism detection; siamese networks; equivariant Vision Transformers; group invariance; contrastive learning; DomainNet.

1 Introduction

Two images depict the same content under a plagiarist’s edits when one differs from the other by a rotation, a reflection, a crop, or a photometric perturbation, and a detector should map both to the same point in feature space. The question we study is whether to enforce this invariance through the encoder architecture or to approximate it through geometric augmentation at training time.

Standard Vision Transformers [1] provide no geometric guarantee. Patch tokenization commutes with neither rotation nor reflection, and learned positional encodings do not extrapolate beyond the training distribution; geometric augmentation can only cover the orbit of a transformation group stochastically, inflates training cost, and makes no promise on transformations unseen during training. Scale alone does not close this gap, since spatial reasoning of large vision models has been shown to fail to improve with parameter count [2]. Geometric deep learning [3–5] formalizes the alternative, in which the desired invariance is built into the architecture as commutation with a group action and then propagates exactly through the network. The siamese template, on which the detector is built, is due to Bromley et al. [6].

Classical near-duplicate pipelines [7] embed geometric invariance directly in the matching procedure and therefore tolerate rotations, reflections and crops by construction, but lose accuracy under domain shift and photometric edits. Modern self-supervised descriptors for image copy detection [8, 9] reverse this profile: they are robust to photometric perturbations and domain shift and define the state of the art on standard copy-detection benchmarks, yet carry no architectural geometric guarantee and depend on augmentation to approximate it. SSCD [8] couples a SimCLR-style contrastive objective on a convolutional backbone with an entropy regularizer that enforces a consistent separation of descriptor vectors across the embedding space and with augmentations that mix pairs of images at training time, and it replaces the standard pooling head with a generalized-mean operator adapted from instance retrieval; the resulting compact descriptor improves over a SimCLR baseline by 48 absolute points of micro average precision on DISC2021. RDCD [9] treats descriptor compactness as a distillation problem: a frozen SSCD-style teacher defines a pairwise similarity structure over an instance queue, and a smaller student is trained to reproduce that structure through a Kullback-Leibler term combined with a hard-negative loss, which yields 64-, 128- and 256-dimensional descriptors that match or exceed the teacher on DISC2021. In both pipelines the robustness to rotation, reflection and crop is sampled rather than guaranteed and remains bounded by the empirical augmentation distribution, which is the gap that an architecturally G -invariant encoder closes by construction. Among architectural alternatives, the Octic ViT [10] guarantees invariance to rotations by multiples of 90 degrees and to reflections, but covers no other angle, while Harmformer [11] extends the guarantee to all rotation angles at a higher compute cost. None of these architectures has been evaluated on pairwise plagiarism detection, which is the specific gap we address.

The contribution is a single siamese pipeline that wraps three encoders, a non-equivariant ViT-Tiny/16 baseline, an Octic ViT, and a Harmformer, and pits architectural group invariance against geometric augmentation under matched conditions. We utilize a controlled augmentation ablation: each equivariant encoder is trained without group augmentation, while the ViT-Tiny/16 baseline is trained with the corresponding group-specific augmentation block, so that any difference in error attributes to architectural inductive bias rather than to data exposure. The computational experiment estimates whether group-invariant encoders trained without the group augmentation block match or exceed the augmented baseline on false positive rate and per-class recall on a balanced cross-domain test of 1 200 pairs sampled from DomainNet [12]; training pairs are constructed on COCO 2017 with the cross-batch protocol of Dorin et al. [13].

The remainder of the paper proceeds as follows. Section 2 reviews siamese detectors, equivariant networks, and equivariant ViTs. Section 3 formalizes the detection task and states the equivariance hypothesis. Section 4 fixes notation for group actions, representations, and equivariance. Section 5 describes the three encoders, the symmetric fusion head, and the loss. Section 6 reports the experimental protocol with placeholder results. Section 7 concludes.

2 Related Work

Twin networks with shared weights and a learned distance were introduced by Bromley et al. [6] for signature verification and have since been adopted across pairwise matching problems. Classical near-duplicate detection combined local descriptors with geometric verification [7], which gave geometric robustness by construction but was brittle under domain shift and photometric edits, where modern visual features are stronger. Dorin et al. [13] applied a siamese ViT with a contrastive loss to image plagiarism on DomainNet and proposed the cross-batch pairing protocol that we reuse without modification; the pipeline reports FPR as the primary metric, but offers no invariance guarantee against geometric perturbations and depends on geometric augmentation for robustness.

Group-equivariant convolutional networks [4] encode discrete symmetries through weight sharing on group orbits, and Weiler and Cesa [5] extend the construction to the continuous Euclidean group $E(2)$ via steerable filters. Bronstein et al. [3] treat equivariance as an inductive bias that improves sample efficiency under group-structured data, and the same line of reasoning motivates the transformer variants used here. The Octic ViT [10] replaces standard transformer layers with D_4 -equivariant ones; by Schur’s lemma, equivariant linear maps decompose blockwise across the five irreducible representations of D_4 , which both constrains parameters and reduces FLOPs by 40% at expressivity comparable to a non-equivariant ViT, but the resulting invariance is restricted to multiples of 90° and reflections. Harmformer [11] replaces patch projections with circular harmonic filters and combines them with self-attention to obtain continuous roto-translational equivariance over $SE(2)$, which

covers every rotation angle but at a higher compute cost than D_4 . Neither architecture has been evaluated on pairwise plagiarism detection, which leaves open whether the geometric inductive bias they provide actually reduces false positives on this specific task.

The contrastive component of our loss follows the formulation of Hadsell et al. [14] and the metric-learning template of Chopra et al. [15]: it depends only on Euclidean distances between embeddings, which means that an orthogonal change of basis on the embedding space leaves it invariant, and that property aligns the loss geometry with the geometry of G -invariant encoders.

3 Problem Statement

We treat image plagiarism detection as a binary classification problem on image pairs: given a pair $(x_1, x_2) \in \mathcal{X} \times \mathcal{X}$, predict $y \in \{0, 1\}$, where $y = 1$ marks x_2 as a plagiarized derivative of x_1 .

Plagiarism is operationalized as the composition of a geometric transformation $g \in G$ with a photometric perturbation $\eta \in \mathcal{P}$, producing $x_2 = (g \circ \eta)(x_1)$.

The geometric component g is drawn from one of the symmetry groups $G \in \{D_4, SE(2)\}$.

D_4 is the dihedral group of the square:

$$D_4 = \{r, s \mid r^4 = s^2 = e, srs = r^{-1}\}, \quad (1)$$

of order eight, where r is the rotation of the image plane by 90° and s is reflection across a fixed axis.

$SE(2)$ is the special Euclidean group of the plane is the semidirect product

$$SE(2) = SO(2) \ltimes \mathbb{R}^2 = \{(R_\alpha, \mathbf{t}) \mid \alpha \in [0, 2\pi), \mathbf{t} \in \mathbb{R}^2\}, \quad (2)$$

acting on coordinates by $(R_\alpha, \mathbf{t}) \cdot \mathbf{x} = R_\alpha \mathbf{x} + \mathbf{t}$.

The photometric component η is drawn from the set

$$\mathcal{P} = \{\text{CJ, GBlur, GN, JPEG, WM}\}, \quad (3)$$

where CJ is brightness and contrast jitter, GBlur is Gaussian blur, GN is additive Gaussian noise, JPEG is JPEG re-compression and WM is a watermark overlay; the set is shared across all training and evaluation regimes.

The dataset is $\mathcal{D} = \{(x_{1,i}, x_{2,i}, y_i)\}_{i=1}^N$ with $x_{\cdot,i} \in \mathcal{X} = [0, 1]^{C \times H \times W}$ and $y_i \in \{0, 1\}$, balanced between positive and negative pairs and stratified by visual domain.

Let $\phi^G, \phi^{\text{aug}}: \mathcal{X} \times \mathcal{X} \rightarrow \{0, 1\}$, where ϕ^G is built around a G -invariant ViT trained without block- G augmentation and ϕ^{aug} around a non-equivariant ViT trained with block- G augmentation. For every transformation $T \in G$ we test

$$\text{FPR}_T(\phi^G) \leq \text{FPR}_T(\phi^{\text{aug}}) \quad \text{and} \quad \text{Recall}_T(\phi^G) \geq \text{Recall}_T(\phi^{\text{aug}}). \quad (4)$$

4 Preliminaries: Equivariance

A left action of a group G on a set $\mathcal{X}$ is a map $G \times \mathcal{X} \rightarrow \mathcal{X}$, $(g, x) \mapsto g \cdot x$, satisfying $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and all $x \in \mathcal{X}$. A linear representation of G on a vector space V is a homomorphism $\rho: G \rightarrow \text{GL}(V)$, so that group multiplication is preserved by $\rho(gh) = \rho(g)\rho(h)$. A subspace $W \subseteq V$ is G -invariant if $\rho(g)W \subseteq W$ for all $g \in G$, and the representation is irreducible when its only G -invariant subspaces are $\{0\}$ and V . A map $\phi: \mathcal{X} \rightarrow \mathbb{R}^d$ is G -equivariant with respect to representations $\rho_{\mathcal{X}}$ on $\mathcal{X}$ and ρ on $\mathbb{R}^d$ if $\phi(\rho_{\mathcal{X}}(g)x) = \rho(g)\phi(x)$ for every $g \in G$ and every $x \in \mathcal{X}$, and it is G -invariant when ρ is the trivial representation, that is, when $\phi(g \cdot x) = \phi(x)$. For plagiarism detection, G -invariance of the encoder ϕ_θ implies that all geometric variants of an image collapse to a single embedding without augmentation, so the head receives identical inputs whether or not a $g \in G$ has been applied; the detection problem reduces to comparing two points whose distance no longer depends on the geometric component of the perturbation.

Lemma 1 (Schur). *Let (V, ρ_V) and (W, ρ_W) be finite-dimensional irreducible representations of a group G over an algebraically closed field, and let $T: V \rightarrow W$ be a G -equivariant linear map, that is,*

$$T \rho_V(g) = \rho_W(g) T \quad \text{for every } g \in G. \quad (5)$$

Then either $T = 0$, or T is an isomorphism. In the latter case V and W are equivalent as representations, and if $V = W$ and $\rho_V = \rho_W$, then $T = \lambda \text{id}_V$ for some scalar λ .

A direct consequence is that any equivariant linear map between fully decomposed representations is block-diagonal across the irreducible decomposition: if $V = \bigoplus_i V_i^{\oplus n_i}$ and $W = \bigoplus_i W_i^{\oplus m_i}$ are decomposed into isotypic components indexed by the irreducibles V_i of G , then a G -equivariant linear map $T: V \rightarrow W$ acts within each isotypic component and cannot mix distinct irreducible types. The Octic ViT [10] relies on this property, since each transformer layer operates independently on each irreducible component of D_4 , so the block-diagonal structure holds by construction and the parameter count drops without loss of expressivity inside each isotypic component.

5 Method

5.1 Siamese pipeline

The detector follows the siamese template of Bromley et al. [6]. The detector is the composition $f_\theta(x_1, x_2) = \sigma(h_\psi(\phi_\theta(x_1), \phi_\theta(x_2)))$, where $\phi_\theta: \mathcal{X} \rightarrow \mathbb{R}^d$ is a shared encoder, h_ψ is a fusion head, and σ is the sigmoid.

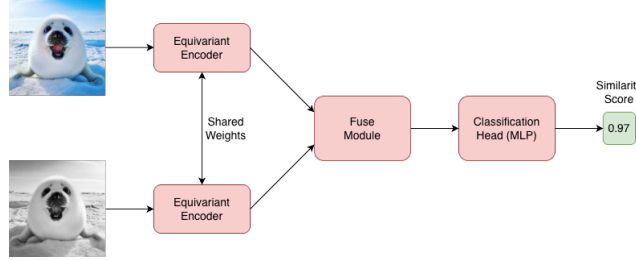


Figure 1: Siamese pipeline with a G -invariant encoder. Both branches share the encoder ϕ_θ , the embeddings z_1, z_2 are passed to the symmetric fusion head h_ψ , and the sigmoid produces the plagiarism probability.

5.2 Encoders

The non-equivariant reference is a ViT-Tiny/16 [1]. A 224×224 input is split into $14 \times 14 = 196$ non-overlapping 16×16 patches, each linearly projected to $\mathbb{R}^{192}$, and the resulting sequence with a prepended [CLS] token is processed by twelve transformer blocks with multi-head self-attention. The [CLS] embedding after the final layer norm is taken as the image representation $z \in \mathbb{R}^{192}$. The encoder has no equivariance guarantee, and at training time it receives both the photometric block and a group-specific geometric block matched to the comparison in which it participates.

The Octic ViT [10] replaces each transformer layer with a D_4 -equivariant variant. By Schur’s lemma the embedding space decomposes across the irreducibles of D_4 ,

$$\mathbb{R}^d \cong (\mathbb{R}^{d/8})_{A_1} \oplus (\mathbb{R}^{d/8})_{A_2} \oplus (\mathbb{R}^{d/8})_{B_1} \oplus (\mathbb{R}^{d/8})_{B_2} \oplus (\mathbb{R}^{d/4})_E, \quad (6)$$

into four one-dimensional isotypic components and one two-dimensional component. D_4 -invariant pooling projects onto the trivial isotypic component A_1 to produce a D_4 -invariant output.

Harmformer [11] replaces patch embeddings with circular harmonic filters that act as steerable feature maps and combines them with self-attention to obtain equivariance under continuous roto-translations.

5.3 Fusion head

The head takes $z_1, z_2 \in \mathbb{R}^d$ and returns a logit through pair features and a two-layer MLP. Set $\delta = |z_1 - z_2|$ for the element-wise absolute difference, $p = z_1 \odot z_2$ for the element-wise product, and $v = [u_\delta(\delta); u_p(p)] \in \mathbb{R}^{2d}$ for the concatenation after independent linear maps; the head value is then $h_\psi(z_1, z_2) = c^\top \text{ReLU}(Av + b)$.

Since $|z_1 - z_2| = |z_2 - z_1|$ and $z_1 \odot z_2 = z_2 \odot z_1$, the head is symmetric in its two arguments.

5.4 Data sampling strategy

Training pairs are constructed with the cross-batch protocol of Dorin et al. [13]. A mini-batch of B source images $\{x^{(i)}\}_{i=1}^B$ is sampled uniformly without replacement from the training pool, and each image is mapped to two branches: an anchor branch $x_1^{(i)}$, on which only the geometry-preserving resize to a fixed input size is applied, and an augmented branch $x_2^{(i)}$, which additionally passes through the photometric block (and, for the non-equivariant baseline, the geometric block of Section 5.6). The pair $(x_1^{(i)}, x_2^{(i)})$ is therefore a clean image and one of its augmented derivatives, which is the unit of positive supervision the detector should map together. Cross-batch pairing then reuses the same two branches across the whole mini-batch: every ordered pair $(x_1^{(i)}, x_2^{(j)})$ with $i, j \in \{1, \dots, B\}$ is included as a training example with label

$$y_{ij} = \mathbb{1}[i = j], \quad (7)$$

so that one mini-batch produces B^2 pairs, of which B are positive ($y_{ii} = 1$) and $B^2 - B$ are negative. The protocol thus emits a label-imbalanced stream with positive-to-negative ratio $1 : (B - 1)$, which motivates the class weights w_+, w_- in (9). To prevent gradient overflow under mixed-precision training we adopt a square-root balance, $w_+/w_- = \sqrt{B - 1}$, with $w_- = 1$.

5.5 Loss function

The total loss function over a labelled dataset $\mathcal{D} = \{(x_{1,i}, x_{2,i}, y_i)\}_{i=1}^N$ is

$$\mathcal{L}_{\text{total}}(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^N \left[\mathcal{L}_{\text{BCE}}(x_{1,i}, x_{2,i}, y_i) + \lambda \mathcal{L}_{\text{contr}}(x_{1,i}, x_{2,i}, y_i) \right], \quad (8)$$

where $\lambda > 0$ is the regularization weight. The binary cross-entropy term acts on the predicted plagiarism probability $\hat{y}_i = f_\theta(x_{1,i}, x_{2,i})$ produced by the detector,

$$\mathcal{L}_{\text{BCE}}(x_1, x_2, y) = -[w_+ y \log \hat{y} + w_- (1 - y) \log(1 - \hat{y})], \quad \hat{y} = f_\theta(x_1, x_2), \quad (9)$$

with class weights $w_+, w_- > 0$ that compensate for the label imbalance induced by the cross-batch pairing of Section 5.4. The contrastive term acts on the encoder embeddings $z_i = \phi_\theta(x_i) \in \mathbb{R}^d$ through the squared L^2 distance,

$$\mathcal{L}_{\text{contr}}(x_1, x_2, y) = \begin{cases} \|\phi_\theta(x_1) - \phi_\theta(x_2)\|_2^2, & \text{if } y = 1, \\ \max(0, m - \|\phi_\theta(x_1) - \phi_\theta(x_2)\|_2)^2, & \text{if } y = 0, \end{cases} \quad (10)$$

with contrastive margin $m > 0$. On a positive pair the term penalizes any non-zero distance between the two embeddings and drives them together; on a negative pair it penalizes embeddings that lie within the margin and drives them apart, vanishing once the L^2 distance reaches m .

5.6 Augmentation blocks

The augmentation pipeline consists of a photometric block applied to all encoders and a geometric block added only to the non-equivariant ViT-Tiny/16 baseline. Inside each block the individual transformations are drawn independently for every pair, and each one is applied with probability $p = 0.3$.

The photometric block draws its elements from the set $\mathcal{P}$ in (3). When sampled, CJ jitters brightness, contrast and saturation with strength 0.3 and hue with strength 0.1, with each channel offset drawn uniformly from the corresponding range; GBlur convolves the image with a 3×3 Gaussian kernel of standard deviation $\sigma \sim \text{Unif}[0.1, 2.0]$; GN adds independent zero-mean Gaussian noise to every pixel with standard deviation $\sigma \sim \text{Unif}[0.01, 0.05]$; JPEG re-encodes the image at integer quality $q \sim \text{Unif}\{10, \dots, 50\}$; and WM overlays a randomly placed text watermark with transparency $\alpha \sim \text{Unif}[0.1, 0.4]$.

The geometric block is matched to the symmetry group of the comparison. In the D_4 comparison it samples, each with probability 0.3, a discrete rotation by $90^\circ k$ with $k \in \{1, 2, 3\}$ drawn uniformly, a horizontal flip and a vertical flip. In the $SE(2)$ comparison it samples, each with probability 0.3, a continuous rotation with angle $\alpha \sim \text{Unif}[0, 2\pi)$ and an isotropic random translation with displacement on each axis drawn uniformly from $\pm 20\%$ of the corresponding image extent.

6 Computational Experiment

The experiment estimates, for each architecture, two figures of merit on a held-out cross-domain test set: the false positive rate FPR and the recall Recall aggregated over the entire set, and the same two metrics stratified by geometric transformation class. The decision threshold is fixed at $\tau = 0.5$, the value used to binarize the sigmoid output of the detector during training, and is not re-tuned on the test set, so test-time metrics reflect the model as deployed. Training hyperparameters are summarized in Table 1.

Table 1: **Training parameters for the models.** Optimizer is AdamW with the listed learning rate and weight decay. It specifies contrastive loss parameters like the regularization parameter λ and margin m , and includes general training parameters such as batch size B and the number of epochs.

Parameter	Learning Rate	Weight Decay	Regularization (λ)	Margin	Batch Size (B)	Epochs
Value	10^{-3}	10^{-2}	0.05	0.5	128	20

Test set construction. The test pool is sampled from DomainNet [12], which spans six visually diverse domains: real, painting, clipart, quickdraw, infograph and sketch. From each domain we pre-select 200 source images, giving 1 200 source images in total. Two pair regimes are built on top of this pool. The *overall* regime forms a single balanced test set of 1 200 pairs, 600 positive and 600 negative (100 positive and 100 negative per domain), with both branches passed through the same training-time augmentation pipeline of Section 5.6 (the photometric block for every model, plus the geometric block matched to the model’s training group), so that the global FPR and Recall are measured on data drawn from the training distribution. The *per-stratum* regime forms one separate test set of 1 200 pairs per geometric stratum of Table 2, again 600 positive and 600 negative, with the photometric block disabled and only the stratum’s geometric transformation applied to the second branch of each positive pair, so that per-stratum Recall isolates the model’s response to that single transformation. No augmentation hyperparameter, no decision threshold and no class weight is changed between training and either evaluation regime.

Stratification by geometric transformation. Per-transformation FPR and Recall are reported on the geometric strata listed in Table 2, chosen to separate exact D_4 -equivariance from continuous $SE(2)$ -equivariance. The strata are: the identity, the three discrete rotations of D_4 , the two reflections of D_4 , two off- D_4 rotations that lie in $SE(2)$ but not in D_4 , and a translation. The identity stratum measures performance under photometric perturbation only and serves as a control; the discrete-rotation and flip strata test D_4 -coverage; the off- D_4 rotations probe continuous-angle behaviour; the translation stratum tests the second factor of $SE(2)$.

Table 2: Geometric strata used for per-transformation FPR and Recall.

Stratum	Geometric transformation g	Probes
Id	identity	photometric-only control
R_{90}	90° rotation	D_4
R_{180}	180° rotation	D_4
R_{270}	270° rotation	D_4
HFlip	horizontal flip	D_4 reflection
VFlip	vertical flip	D_4 reflection
R_{45}	45° rotation	$SE(2) \setminus D_4$
R_{135}	135° rotation	$SE(2) \setminus D_4$
$T_{0.2}$	translation by $\text{Unif}([-0.2, 0.2]^2)$	$SE(2)$ translation

Results. Table 3 reports, for each of the five trained models, the overall FPR and Recall on the 1 200-pair test with training-time augmentation, and the per-stratum Recall on each geometric stratum of Table 2, all at the fixed threshold $\tau = 0.5$. Rows are grouped into a common photometric-only baseline and the two per- G comparisons. The photometric-only ViT-Tiny baseline behaves as a control: overall Recall is 99.83% because the overall regime applies only the photometric block to its second branch, but per-stratum Recall drops to 11–31% on the D_4 orbit and collapses to $\leq 1.2\%$

Table 3: **Quantitative performance comparison on DomainNet test subsets across model architectures.** All metrics are evaluated at a fixed decision threshold of $\tau = 0.5$. The first two metric columns come from the overall regime (one 1 200-pair test with the training-time augmentation of each model on both branches): FPR is the global false positive rate, Recall is the global recall on the same set. The remaining columns come from the per-stratum regime (one 1 200-pair test per geometric stratum, photometric block off, only the stratum’s geometric transform applied to the positive view). Stratum abbreviations: R_α – rotation by α° , HFlip / VFlip – horizontal / vertical flip, $T_{0.2}$ – random translation up to $\pm 20\%$ on each axis. Best FPR in bold.

Model	Overall		Per-stratum Recall %							
	FPR %	Recall %	R_{90}	R_{180}	R_{270}	HFlip	VFlip	R_{45}	R_{135}	$T_{0.2}$
ViT-Tiny/16 (photometric only)	7.67	99.83	11.67	16.50	10.67	31.33	21.17	0.50	0.33	1.17
ViT-Tiny/16 + block G_{D_4}	3.67	100.00	100.00	100.00	100.00	100.00	100.00	0.33	0.33	1.00
Octic ViT	1.17	99.83	100.00	100.00	100.00	100.00	100.00	0.33	0.33	0.50
ViT-Tiny/16 + block $G_{SE(2)}$	16.17	95.83	100.00	100.00	100.00	100.00	100.00	31.00	30.67	99.17
Harmformer	12.00	99.83	100.00	100.00	100.00	94.17	94.17	0.67	0.67	2.50

on R_{45} , R_{135} and $T_{0.2}$, which confirms that the encoder by itself carries no geometric robustness. In the D_4 comparison the augmented ViT-Tiny baseline and the Octic ViT both reach 100% Recall on every D_4 stratum, so the architectural D_4 -invariance fully replaces the corresponding geometric augmentation block; and at the same time the overall FPR of the Octic ViT is 1.17% versus 3.67% for the augmented ViT-Tiny, a 2.5 percentage-point improvement at matched Recall, so the architectural inductive bias does not only match augmentation on the orbit, it more than triples the precision of the augmented baseline at fixed threshold. Both models still fail on the off-orbit strata R_{45} , R_{135} and $T_{0.2}$ ($\leq 1\%$ Recall), as expected for an encoder whose invariance is restricted to D_4 . In the $SE(2)$ comparison the augmented ViT-Tiny reaches 100% on the D_4 strata and 99.17% on $T_{0.2}$ but only 31.00% and 30.67% on R_{45} and R_{135} , with the highest overall FPR in the table (16.17%) and overall Recall that drops to 95.83%, indicating that the continuous-rotation augmentation distribution under-samples off-orbit angles and adds noise to the negative side. The Harmformer trained without geometric augmentation reaches 100% on the discrete D_4 rotations and 94.17% on the two flips, but $\leq 0.67\%$ on R_{45}/R_{135} and 2.50% on $T_{0.2}$, with overall FPR of 12.00% and overall Recall of 99.83%.

Discussion. The two comparisons return different verdicts on the equivariance hypothesis H_1 . For $G = D_4$, H_1 is supported in both directions: the architecturally D_4 -invariant Octic ViT matches the augmented baseline on Recall along every D_4 stratum and improves FPR by 2.5 percentage points (and overall precision from $1/(1 + 0.0367)$ to $1/(1 + 0.0117)$), so the geometric augmentation block can be dropped without loss of detection performance, exactly the experimental claim of Section 5.6. For $G = SE(2)$, the verdict is mixed: the Harmformer beats the augmented ViT-Tiny on overall FPR (12.00% vs 16.17%) and on overall Recall (99.83% vs 95.83%) and reproduces the discrete D_4 orbit at 100%, yet collapses to under 1% on the off- D_4 angles R_{45} and R_{135} and to 2.5% on translation $T_{0.2}$, which are exactly the strata its architecture is supposed to cover, while the augmented ViT-Tiny reaches 31% on R_{45}/R_{135} and 99% on $T_{0.2}$. The architectural advantage on overall metrics therefore does not come from continuous $SE(2)$ -equivariance but from the absence of the noisy continuous-rotation augmentation that drags the ViT – Tiny + $G_{SE(2)}$ baseline towards the decision threshold. The most likely cause of the off- D_4 collapse of the Harmformer is the small input resolution: the released code resizes images to 32×32 before the encoder, and the circular harmonic filters rely on continuous rotation acting on a sufficiently sampled pixel grid; at 32×32 , bilinear interpolation around non-axis-aligned angles introduces enough discretization noise to break the analytic $SE(2)$ -equivariance, which is consistent with R_{45}/R_{135} Recall collapsing while the discrete D_4 orbit, where rotation acts by axis-aligned permutations of pixels, is unaffected. The augmented ViT-Tiny shows the symmetric failure on the same strata (31% Recall on R_{45}/R_{135} despite continuous-angle training), which suggests that the augmentation distribution at 32×32 is also not dense enough to cover $SO(2)$ effectively. Both observations point to the input resolution rather than to the equivariance principle as the binding constraint for the $SE(2)$ comparison; the D_4 result, by contrast, is robust to this issue because D_4 incurs no interpolation cost, which is the regime in which architectural G -invariance dominates.

7 Conclusion

We compared architectural group invariance against geometric augmentation in a single siamese plagiarism detector, on the dihedral group of the square and the special Euclidean group of the plane. For the dihedral group the architecturally invariant encoder matches the augmented baseline on every right-angle rotation and reflection and improves precision on negatives, which supports the equivariance hypothesis and lets the geometric augmentation block be dropped without loss. For the special Euclidean group the verdict is mixed: the continuously equivariant encoder improves overall metrics and is exact on the discrete dihedral orbit, but fails on off-axis rotations and on translation, which we attribute to interpolation noise at low input resolution rather than to the equivariance principle. Future work will rerun the comparison at higher input resolution, so that arbitrary rotation angles act on enough pixels for the architectural invariance of the equivariant encoders to hold in practice, and broaden the set of tested transformations beyond rotations, reflections and translation to also cover scale and projective perturbations.

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